# BCB744 Biostatistics self-assessment questions and answers

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This document gathers the self-assessment questions and model answers from the biostatistics chapters. Answers are shown by default here. The chapter versions retain only the questions.

## 2. Summarising biological data

! Self-assessment Task 2-1

How would you manually calculate the mean value for the `normal_data` we generated in the lecture? **(/3)**

**Answer**

```
set.seed(666)
n <- 5000
mean <- 0
sd <- 1
normal_data <- rnorm(n, mean, sd)
round(sum(normal_data) / length(normal_data), 3)
```

```
[1] 0.009
```

- ✓✓✓ The mean value of the `normal_data` is calculated by summing all the data points and dividing by the number of data points. Assign three marks if they correctly calculate the mean of the `normal_data` without using the function `mean()`.

! Self-assessment Task 2-2

Find the `faithful` dataset and describe both variables in terms of their measures of central tendency. Include graphs in support of your answers (use `ggplot()`), and conclude with a brief statement about the data distribution. **(/10)**

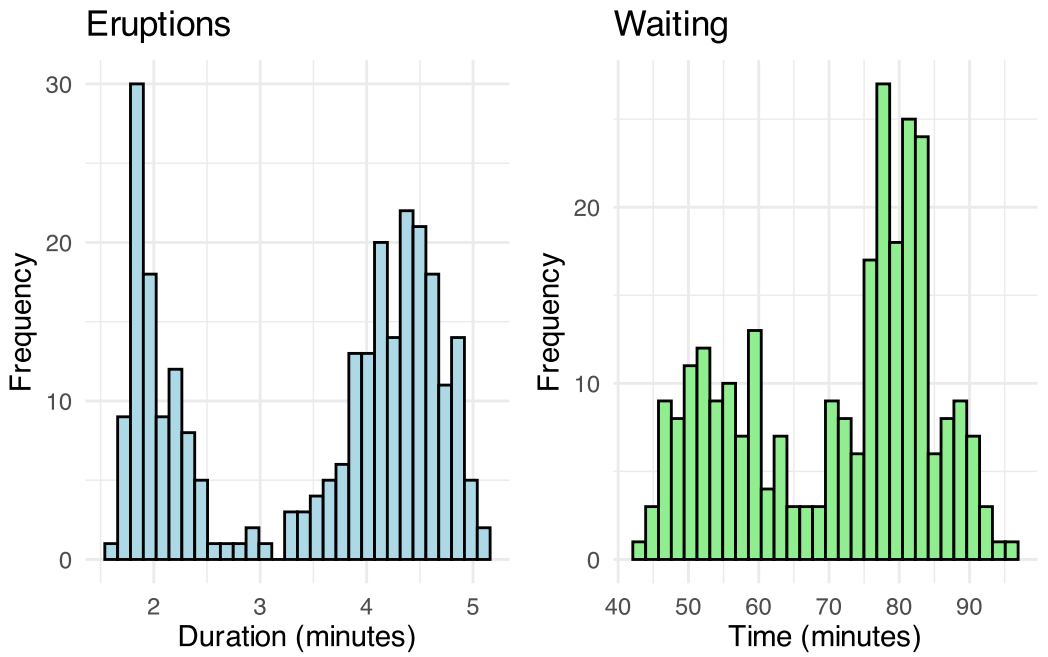
**Answer**

```
# Load the faithful dataset
data(faithful)
library(ggpubr)

plt1 <- ggplot(faithful, aes(x = eruptions)) +
  geom_histogram(fill = "lightblue", colour = "black") +
  labs(title = "Eruptions", x = "Duration (minutes)", y = "Frequency") +
  theme_minimal()

plt2 <- ggplot(faithful, aes(x = waiting)) +
  geom_histogram(fill = "lightgreen", colour = "black") +
  labs(title = "Waiting", x = "Time (minutes)", y = "Frequency") +
  theme_minimal()

# Arrange the plots side by side
ggarrange(plt1, plt2, ncol = 2)
```




```
# Central tendency measures
mean(faithful$eruptions)
median(faithful$eruptions)
kurtosis(faithful$eruptions)
```

*2. Waiting Time*

*3. Conclusion About the Data's Distribution*

Due to the bimodal nature of the measured variables, neither conforms to expectations of central tendency characterisation. Yes, we can compute mean, median values but their

! Self-assessment Task 2-3

Manually calculate the variance and SD for the `normal_data` we generated in the lecture. Make sure your answer is the same as those reported there. **(/5)**

**Answer**

```
# Using the equations for sample variance and sd
# (not the built-in functions), do:

# Variance
(norm_var <- round(sum((normal_data - mean(normal_data))^2) /
                     (length(normal_data) - 1), 3))
```

```
[1] 1.002
```

```
# Standard deviation
(norm_sd <- round(sqrt(norm_var), 3))
```

```
[1] 1.001
```

! Self-assessment Task 2-4

Write a few lines of code to demonstrate that the $(0 - 0.25]$, $(0.25 - 0.5]$, $(0.5 - 0.75]$, $(0.75 - 1]$ quantiles of the `normal_data` we generated in the lecture indeed conform to the formal definition for what quantiles are. I.e., show manually how you can determine that 25% of the observations indeed fall below −0.66 for the `normal_data`. Explain the rationale to your approach. **(/10)**

**Answer**

```
# Generate random data from a normal distribution
set.seed(666)
n <- 5000 # Number of data points
mean <- 0
sd <- 1
normal_data <- rnorm(n, mean, sd)

# Calculate the quantiles
q_25 <- quantile(normal_data, p = 0.25)
q_50 <- quantile(normal_data, p = 0.50)
q_75 <- quantile(normal_data, p = 0.75)
q_100 <- quantile(normal_data, p = 1.00)

# Verify each quantile interval
(count_0_to_25 <- sum(normal_data <= q_25))
(count_25_to_50 <- sum(normal_data > q_25 & normal_data <= q_50))
(count_50_to_75 <- sum(normal_data > q_50 & normal_data <= q_75))
(count_75_to_100 <- sum(normal_data > q_75 & normal_data <= q_100))

# Calculate percentages
(perc_0_to_25 <- count_0_to_25 / n * 100)
(perc_25_to_50 <- count_25_to_50 / n * 100)
(perc_50_to_75 <- count_50_to_75 / n * 100)
(perc_75_to_100 <- count_75_to_100 / n * 100)

# Make a figure to visualise
library(ggplot2)
ggplot(data.frame(x = normal_data), aes(x = x)) +
  geom_histogram(binwidth = 0.5, fill = "lightblue", colour = "black") +
   geom_vline(xintercept = c(q_25, q_50, q_75), colour = c("red", "blue",
"green"), linetype = "dashed") +
  labs(title = "Histogram of normal_data with quantiles", x = "Value", y =
"Frequency") +
  theme_minimal()
```

- ☒ (xx/10) Take that the correctly calculate the quantiles and verify that approximately 25% of the observations fall within each quantile interval. The code should also provide a clear explanation of the approach. The formal definition states that the $p$-th quantile is a value $q_p$ such that the proportion of observations less than or equal to $q_p$ is approximately $p$.

*Approach*

! Self-assessment Task 2-5

a. Explain the output of `dimnames()` when applied to the `penguins` dataset. **(/2)**
b. Explain the output of `str()` when applied to the `penguins` dataset. **(/3)**

**Answer**

```
library(palmerpenguins)
data(penguins)

# a.
dimnames(penguins)
```

```
[[1]]
  [1] "1"   "2"   "3"   "4"   "5"   "6"   "7"   "8"   "9"   "10"  "11"  "12"
 [13] "13"  "14"  "15"  "16"  "17"  "18"  "19"  "20"  "21"  "22"  "23"  "24"
 [25] "25"  "26"  "27"  "28"  "29"  "30"  "31"  "32"  "33"  "34"  "35"  "36"
 [37] "37"  "38"  "39"  "40"  "41"  "42"  "43"  "44"  "45"  "46"  "47"  "48"
 [49] "49"  "50"  "51"  "52"  "53"  "54"  "55"  "56"  "57"  "58"  "59"  "60"
 [61] "61"  "62"  "63"  "64"  "65"  "66"  "67"  "68"  "69"  "70"  "71"  "72"
 [73] "73"  "74"  "75"  "76"  "77"  "78"  "79"  "80"  "81"  "82"  "83"  "84"
 [85] "85"  "86"  "87"  "88"  "89"  "90"  "91"  "92"  "93"  "94"  "95"  "96"
 [97] "97"  "98"  "99"  "100" "101" "102" "103" "104" "105" "106" "107" "108"
[109] "109" "110" "111" "112" "113" "114" "115" "116" "117" "118" "119" "120"
[121] "121" "122" "123" "124" "125" "126" "127" "128" "129" "130" "131" "132"
[133] "133" "134" "135" "136" "137" "138" "139" "140" "141" "142" "143" "144"
[145] "145" "146" "147" "148" "149" "150" "151" "152" "153" "154" "155" "156"
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[265] "265" "266" "267" "268" "269" "270" "271" "272" "273" "274" "275" "276"
[277] "277" "278" "279" "280" "281" "282" "283" "284" "285" "286" "287" "288"
[289] "289" "290" "291" "292" "293" "294" "295" "296" "297" "298" "299" "300"
[301] "301" "302" "303" "304" "305" "306" "307" "308" "309" "310" "311" "312"
[313] "313" "314" "315" "316" "317" "318" "319" "320" "321" "322" "323" "324"
[325] "325" "326" "327" "328" "329" "330" "331" "332" "333" "334" "335" "336"
[337] "337" "338" "339" "340" "341" "342" "343" "344"
```

- ☒ `dimnames()` returns the names of the rows and columns of the dataset.
- ☒ The rows are numbered 1 to 344, and the columns are named `species`, `island`, `bill_length_mm`, `bill_depth_mm`, `flipper_length_mm`, `body_mass_g`, `sex`, `year`.
- ☒ `str()` provides a concise summary of the dataset's structure. It shows the number of observations (344) and variables (8), and their data types.
- ☒ The dataset contains 344 observations and 8 variables.
- ☒ The variables are a mix of character, factor, and numeric data types.

```
#tibble [344 × 8] (S3: tbl_df/tbl/data.frame)
$ species          : Factor w/ 3 levels "Adelie","Chinstrap",..: 1 1 1
```

! Self-assessment Task 2-6

Explain the output of `summary()` when applied to the `penguins` dataset. **(/3)**

**Answer**

```
summary(penguins)
```

- The `summary()` function provides a concise summary (obviously) of the dataset's numerical variables. It includes the minimum, 1st quartile, median, mean, 3rd quartile, and maximum values for each variable (*i.e.* central tendency, some view of the dispersion). For categorical variables and it shows the frequency of each level. We caqn gain some basic insight into the distribution of the data and identify potential outliers or missing values.
- ☒ (x 3) Assign a few marks if they correctly describe the output of `summary()` with specific reference to the `penguins` dataset. For example:
  - We can see that the penguin's bill depth (in mm) is close to normally distributed given that the mean value of 17.15 mm is very close to the median of 17.30 mm, with the minimum, maximum values at 13.10 mm and 21.50 mm and respectively. There are two missing values here. The bill length, however, is slightly skewed to the right with a mean of 43.92 mm, a median of 44.45 mm…

! Self-assessment Task 2-7

Why is it important to consider the grouping structures that might be present within our datasets? **(/2)**

**Answer**

Simpson's Paradox: Relationships seen in grouped data can reverse/disappear when examining subgroups separately.

Heterogeneity: Different groups within data often exhibit distinct statistical properties. A a single mean (and/or SD) across all observations masks this variability and can provide a distorted view of the real phenomena.

Statistical Independence: Many statistical tests assume independence of observations. If data contain hierarchical or nested structures (*e.g.*, individuals within ecosystems, repeated measurements from the same subjects), this assumption is violated, might invalidate statistical inferences.

Group-Specific Insights: Analysing data by relevant groupings (*e.g.* a population of a flowering plant in different climatic regions or time periods) may show relevant patterns and differences that would otherwise remain hidden in aggregate statistics.

So, if we do not account for group structures, summary statistics may reflect artificial central tendencies, dispersions that do not realistically represent any meaningful subpopulation within the data. This limits the validity and usefulness of the analysis.

- ☒ (x 2) … give some marks if some if what is above is mentioned.

# 3. Visualising data

! Self-assessment Task 3-1

a. Using a tidy workflow, assemble a summary table of the **palmerpenguins** dataset that has a similar appearance as that produced by `psych::describe(penguins)`. **(/5)**

   - For bonus marks (which will not count anything), apply a beautiful, creative styling to the table using the **kable** package. Try and make it as publication ready as possible. Refer to a few journal articles to see how to professionally typeset tables.

b. Still using the **palmerpenguins** dataset, perform an exploratory data analysis to investigate the relationship between penguin species, their morphological traits (bill length and bill depth, flipper length, and body mass). Employ the tidyverse approaches learned earlier in the module to explore the data, account for the grouping structures present within the dataset. **(/10)**

c. Provide visualisations (use Figure 4 as inspiration) and summary statistics to support your findings and elaborate on any observed patterns or trends. **(/10)**

d. Ensure your presentation is professional and adhere to the standards required by scientific publications. State the major aims of your analysis and the patterns you seek. Using the combined findings from the EDA and the figures produced here, discuss the findings in a formal Results section. **(/5)**

**Answer**

**a. (/5)**

A tidy workflow selects numeric columns, pivots to long form, and summarises across all variables simultaneously. The resulting table should include the same statistics that `psych::describe()` reports: number of observations, mean, standard deviation, median, minimum, maximum, range, skewness, and kurtosis.

- ☒ Only numeric variables are selected before summarising. (1)
- ☒ At least *n*, mean, and SD are computed for each variable. (1)
- ☒ Median, min, and max are also included, giving a table with comparable scope to `psych::describe()`. (1)
- ☒ Output is passed to a table-formatting function (`kable()` or equivalent) rather than printed as raw R output. (1)
- ☒ The table is readable: columns are labelled, values are rounded to a sensible number of decimal places. (1)

```
library(palmerpenguins)

penguins_summary <- penguins |>
  select(where(is.numeric)) |>
  pivot_longer(everything(), names_to = "variable", values_to = "value") |>
```

The summary table and scatter plot together reveal the major structure in the data. *Gentoo* penguins are the largest species by both body mass (mean ≈ 5092 g) and flipper length (mean ≈ 217 mm). *Adélie* and *Chinstrap* penguins are similar in overall size but differ clearly in bill dimensions: *Adélie* bills are shorter and deeper, while *Chinstrap* bills are longer and shallower. The bill length–depth scatter plot illustrates a within-species positive relationship that is obscured in the pooled data by a between-species negative association — a pattern worth noting because the other two species by approximately 1000 g and 20 mm respectively.

**b. (/10)**

We characterised variation in bill length, bill depth, flipper length, and body mass across three penguin species using grouped summaries and visualisations. Adélie and Chinstrap penguins are similar in overall size but differ in bill dimensions, with Chinstrap birds having longer, shallower bills. Gentoo penguins are the largest by both body mass (mean ≈ 5092 g) and flipper length (217 ± 7 mm), exceeding the other two species by approximately 1000 g and 20 mm respectively.

# 7. *t*-tests

! Self-assessment Task 7-1

Please refer to this two-sided two-sample *t*-test:

```
# random normal data
set.seed(666)
r_two <- data.frame(dat = c(rnorm(n = 20, mean = 4, sd = 1),
                            rnorm(n = 20, mean = 5, sd = 1)),
                    sample = c(rep("A", 20), rep("B", 20)))

# perform t-test
# note how we set the `var.equal` argument to TRUE because we know
# our data has the same SD (they are simulated as such!)
t.test(dat ~ sample, data = r_two, var.equal = TRUE)
```

a. Repeat this analyses using the Welch's `t.test()`. **(/5)**
b. Repeat your analysis, above, using the even more old-fashioned Equation 4 in the lecture. Show the code, talk us through the step you followed to read the *p*-values off the table of *t*-statistics. **(/10)**

**Answer**

**a. (/5)**

Welch's *t*-test differs from Student's *t*-test in one key respect: it does not assume equal variances between the two groups. In R, the only change required is setting `var.equal = FALSE`.

- ☒ Data are recreated with the same seed and parameters as the original code block. (1)
- ☒ `var.equal = FALSE` is set correctly in the `t.test()` call. (1)
- ☒ The output is read: *t*-statistic, degrees of freedom, and *p*-value are correctly identified. (1)
- ☒ The degrees of freedom from Welch's test are non-integer (the Satterthwaite approximation), which differs from Student's test where df = $n_A + n_B - 2 = 38$. This difference is noted. (1)
- ☒ A correct conclusion is stated: the *p*-value is below 0.05, so the null hypothesis of equal means is rejected at the 5% significance level. (1)

```
set.seed(666)
r_two <- data.frame(dat = c(rnorm(n = 20, mean = 4, sd = 1),
                            rnorm(n = 20, mean = 5, sd = 1)),
                    sample = c(rep("A", 20), rep("B", 20)))

# Welch's t-test: var.equal = FALSE
t.test(dat ~ sample, data = r_two, var.equal = FALSE)
```

The *t*-statistic and *p*-value are nearly identical to the Student's test in this case because the sample variances happen to be close, but the degrees of freedom differ. Welch's test produces a non-integer df (here around 37.9) via the Welch–Satterthwaite equation, whereas Student's test gives an integer df of 38.

**b. (/10)**

To read the *p*-value from a printed *t*-table: locate the row for the degrees of freedom (round **down** to the nearest integer if the df is non-integer, which is conservative), then scan across the columns for the critical values at different significance levels. If $|t_{\text{calc}}|$ exceeds the tabulated critical value for $\alpha = 0.05$ (two-tailed), reject $H_0$. Here the critical value at df = 37 and $\alpha = 0.05$ is approximately 2.026; the computed $|t|$ exceeds this, so $H_0$ is rejected.

```
xA <- r_two$dat[r_two$sample == "A"]
xB <- r_two$dat[r_two$sample == "B"]

# t-statistic
t_stat <- (mean(xA) - mean(xB)) /
  sqrt(var(xA) / length(xA) + var(xB) / length(xB))
```

The Welch *t*-statistic for the difference of two means is

! Self-assessment Task 7-2

For each scenario below, identify the correct *t*-test design (one-sample, two-sample, or paired) and state which assumption needs to be checked, and where:

a. You measure the resting heart rates of 25 athletes and compare them with the published population mean of 72 bpm. **(/2)**
b. You compare the nitrogen content of leaves from trees in two adjacent forest types (20 trees per forest). **(/2)**
c. You weigh 15 juvenile tortoises in March and again in October and ask whether they gained mass. **(/2)**
d. You measure the length of left and right tibiae from 18 rabbits. **(/2)**

Discuss with a partner whether (b) and (d) are structurally similar or different. What changes between them? **(/2)**

**Answer**

**a. (/2)** You measure resting heart rates of 25 athletes and compare with the published population mean of 72 bpm.

- ☒ **One-sample *t*-test.** There is a single sample of measurements and a known reference value (the published population mean); no second group is sampled. (1)
- ☒ **Assumption to check:** approximate normality of the 25 heart-rate measurements (histogram or Q-Q plot). With $n = 25$ the Central Limit Theorem provides modest protection, but a strongly skewed distribution would warrant a transformation or a non-parametric alternative. (1)

---

**b. (/2)** You compare nitrogen content of leaves from trees in two adjacent forest types (20 trees per forest).

- ☒ **Two-sample (independent-samples) *t*-test.** Trees in the two forests are distinct, unrelated individuals — there is no natural pairing between a tree in Forest 1 and a tree in Forest 2. (1)
- ☒ **Assumptions to check:** approximate normality within each group (histogram or Q-Q plot per forest), and homogeneity of variance (Levene's test or an *F*-test; alternatively, use Welch's *t*-test to avoid the equal-variance assumption). Independence of observations within each forest is also required — trees sampled in a cluster may not be independent. (1)

---

**c. (/2)** You weigh 15 juvenile tortoises in March and again in October to ask whether they gained mass.

- ☒ **Paired *t*-test.** The same 15 individuals are measured twice; each March measurement is naturally paired with the October measurement for the same tortoise. The analysis is performed on the 15 *differences* (October − March), not on the raw measurements. (1)
- ☒ **Assumption to check:** approximate normality of the **differences**, not of the raw measurements separately. With only 15 differences, the assumption is more sensitive to outliers than in larger samples. (1)

**d. (/2)** You measure the length of left and right tibiae from 18 rabbits.

- ☒ **Paired *t*-test.** Each rabbit contributes one left and one right measurement — the same individual appears in both groups, with equal sample sizes ($n = 18$ pairs). (1)
- ☒ **Assumption to check:** approximate normality of the **differences** (left − right) within each rabbit. This is the quantity the test actually operates on; the raw left and right lengths may each be non-normal if rabbits differ substantially in overall size, but their within-rabbit differences are typically much more tightly distributed. (1)

**Discussion: (b) and (d) structurally similar or different?**

Both (b) and (d) involve two groups with equal sample sizes ($n = 20$ trees per forest; $n = 18$ rabbits), and in both cases the analysis is performed on the differences between the two measurements (left − right tibia; Forest 1 − Forest 2). In (d), the left and right tibia from the same individual rabbit are naturally paired, because they share the same genetic background, age, and overall body size. In (b), the trees in Forest 1 and Forest 2 are distinct individuals with no natural pairing, so the two groups are independent. The structural difference between (b) and (d) is therefore whether the two measurements are linked by a shared individual.

# 8. ANOVA

! Self-assessment Task 8-1

a. Why should we not just apply *t*-tests once per each of the pairs of comparisons we want to make? **(/3)**

b. Using the table above, calculate the probability of at least one false positive when comparing five groups ($K = 5$) at $\alpha = 0.05$ using pairwise *t*-tests. How many comparisons would that involve? Then calculate the same probability at $\alpha = 0.01$. Is a Bonferroni correction or a single ANOVA the better solution? Discuss with a partner and be ready to explain your reasoning. **(/5)**

**Answer**

a. Applying *t*-tests repeatedly across multiple pairwise comparisons increases the probability of committing at least one Type I error — that is, falsely rejecting a true null hypothesis. This inflation of the collective error rate arises because each test is conducted at a fixed significance level (*e.g.*, $\alpha = 0.05$), but the cumulative chance of error grows with the number of tests. So, the overall inference becomes unreliable. Instead, methods such as ANOVA, adjusted p-values (*e.g.* or Bonferroni correction) should be used to control for this multiplicity.

- ☒ Identifying the problem of multiple comparisons.
- ☒ Explaining the consequence: increased risk of Type I error.
- ☒ Suggesting a solution: using ANOVA or adjusted p-values.

b.

```
# Number of pairwise comparisons for K groups
K <- 5
comparisons <- K * (K - 1) / 2
prob_at_least_one_false_pos <- 1 - (1 - 0.05)^comparisons
cat("Comparisons:", comparisons, "\nP(at least 1 Type I error):",
    round(prob_at_least_one_false_pos, 3), "\n")
```

! Self-assessment Task 8-2

a. What does the outcome say about the chicken masses? Which ones are different from each other? **(/2)**
b. Devise a graphical display of this outcome. **(/4)**

**Answer**

a. **Interpretation of ANOVA Output**
   - ☒ (x 2) The one-way ANOVA tests the null hypothesis that all four diet groups have the same mean chicken mass at Day 21. The output shows an $F$-statistic of 4.655 and a $p$-value of < 0.05. Since this $p$-value is less than the conventional 0.05 threshold, we reject the null hypothesis, conclude that there is evidence of a statistically significant difference in mean chicken mass between at least two of the diet groups.
   - Before accepting the ANOVA result, I should check whether the grouped summaries and the residual diagnostics support approximate normality and similar variance. If those assumptions fail badly, the usual ANOVA result may not be reliable.
   - Assuming the assumptions are met, note that the ANOVA alone does not identify which specific diets differ. To determine that, a *post hoc* comparison — such as Tukey's HSD test — is needed. Without such pairwise analysis, we cannot yet say which diets are different from one another. A graph may help visualise these differences.
b. **Graphical Display**
   - ☒ (x 4) A suitable figure would be a boxplot of chicken weight at Day 21, stratified by Diet. This graph shows the distribution, central tendency, and spread for each diet group, makes it easy to infer where potential differences may lie. I have created two options: A) showing also the mean ± CI and B) showing the raw data points.
   - Below, we see that Diet 3's median is visibly higher, its interquartile range (or CI) does not overlap with Diet 1. This suggests a substantive difference. However and the plot should be interpreted alongside formal statistical comparisons (the ANOVA) to avoid misreading potential noise as a signal.
   - You could also have done a boxplot showing the mean ± SD, but this is less informative than the boxplot with the mean ± CI.

```
library(tidyverse)
library(ggpubr)

# Filter the data for Day 21
chicks <- as_tibble(ChickWeight)
chicks_day21 <- chicks %>%
  filter(Time == 21)

# Create the plot
plt1 <- ggplot(chicks_day21, aes(x = Diet, y = weight, fill = Diet)) +
  geom_boxplot(alpha = 1.0, outlier.shape = NA) +
  stat_summary(fun = mean, geom = "point", shape = 20, size = 3, colour =
"black") +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2,
colour = "black") +
  theme_minimal(base_size = 14) +
  labs(
```

! Self-assessment Task 8-3

Look at the help file for the `TukeyHSD()` function to better understand what the output means.

a. How does one interpret the results? What does this tell us about the effect that that different diets has on the chicken weights at Day 21? **(/3)**
b. Figure out a way to plot the Tukey HSD outcomes in **ggplot**. **(/10)**
c. Why does the ANOVA return a significant result, but the Tukey test shows that not all of the groups are significantly different from one another? **(/3)**

**Answer**

a. **Interpretation of Tukey HSD**
   - ☒ (x 3) The Tukey HSD test, below, compares all possible pairs of means to determine which specific groups are different. The output shows the differences in means between each pair of diets, along with the associated confidence intervals, *p*-values. A significant difference is indicated by a *p*-value less than 0.05 and a confidence interval that does not include zero.
   - ☒ (x 3) The Tukey HSD test results indicate that Diet 1 and Diet 3 are significantly different from each other, as the confidence interval does not include zero, the *p*-value is less than 0.05. This suggests that Diet 3 leads to a higher mean chicken mass compared to Diet 1. The other diet comparisons do not show significant differences.

```
chicks.aov1 <- aov(weight ~ Diet, data = filter(chicks, Time == 21))
TukeyHSD(chicks.aov1)
```

b. **Tukey HSD outcomes presented with `ggplot()`.**
   - ☒ (x 10) One way to do it is like this:

```
# Create a data frame from the Tukey HSD results
tukey_results <- as.data.frame(TukeyHSD(chicks.aov1)$Diet)
tukey_results <- tukey_results %>%
  rownames_to_column(var = "Comparison")

# Create the plot
ggplot(tukey_results, aes(x = Comparison, y = diff)) +
  geom_errorbar(aes(ymin = lwr, ymax = upr), width = 0.2) +
  geom_point(aes(fill = `p adj` < 0.05), size = 3,
             colour = "black", shape = 21) +
  scale_fill_manual(values = c("deepskyblue2", "deeppink2"),
                    labels = c("Not Significant", "Significant")) +
  labs(
    x = "Diet Comparison",
    y = "Difference in Means",
    title = "Tukey HSD Results"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")
```

c. **Why the ANOVA is significant but Tukey is not**
   - ☒ (x 3) The ANOVA tests whether there is at least one significant difference among the groups, while the Tukey HSD test specifically identifies which groups are different. The ANOVA can be significant even if not all groups are different, as it only requires one group to differ from the others. In this case, the ANOVA was significant, but the Tukey test showed that not all groups were significantly different from each other, indicating

! Self-assessment Task 8-4

a. What do you conclude from the above series of ANOVAs? **(/3)**
b. What problem is associated with running multiple tests in the way that we have done here? **(/2)**

**Answer**

a. **Conclusions from the series of ANOVAs**
   - ☒ (x 3) The sequential ANOVAs across times 0, 2, 10, and 21 reveal how the effect of diet on chicken mass develops over time. At time 0, there is no evidence of a difference between diet groups ($p > 0.05$), which is expected, as all chicks begin from a similar baseline before dietary interventions have had time to act. By time 2, a statistically significant effect appears ($p < 0.05$), and this effect becomes more pronounced at time 10 ($p < 0.001$). By time 21, the effect remains significant ($p < 0.005$), though the $F$-statistic is slightly lower than at time 10. This temporal pattern suggests that dietary effects on body mass begin to diverge early, become more detectable over time. The implication is that there is a growing differentiation in growth trajectories due to diet.
b. **Problems with running multiple tests**
   - ☒ (x 2) Running multiple ANOVAs increases the risk of Type I error, as each test has a chance of falsely rejecting the null hypothesis. This inflation of the error rate can lead to erroneous conclusions about the significance of results. To mitigate this, we should consider using a single ANOVA model that includes time as a factor, rather than running separate tests for each time point. Or, we may use a regression model that includes time as a continuous variable.

! Self-assessment Task 8-5

a. How is time having an effect? **(/3)**
b. What hypotheses can we construct around time? **(/2)**

**Answer**

a. **Time's effect**
   - ☒ The effect of time is should be taken as an important influence (independent variable) in the experimental design, because ...
   - ☒ ... we expect the mean chicken mass changes over time as the chickens eat and grow.
   - ☒ Since time is experimentally important, it should be considered in the hypothesis that informs the design of the ANOVA model.
b. **Hypotheses around time**
   - ☒ The null hypothesis is that there is no difference in mean chicken mass across different time points.
   - ☒ The alternative hypothesis is that there is a difference in mean chicken mass across different time points.

! Self-assessment Task 8-6

a. Write out the hypotheses for this ANOVA. **(/2)**
b. What do you conclude from the above ANOVA? **(/3)**

**Answer**

a. **Hypotheses for the ANOVA**
   - ☒ Null Hypothesis ($H_0$): The mean chicken mass is the same at all time points; that is, time has no effect on mass. Formally we would write this as: $\mu_0 = \mu_2 = \mu_{10} = \mu_{21}$.
   - ☒ Alternative Hypothesis ($H_1$): At least one time point has a mean chick mass that differs from the others; time has an effect on chick mass.
b. **Conclusions from the ANOVA**
   - ☒ (x 3) The ANOVA output reports an $F$-statistic of 234.8 with a $p$-value far $< 0.001$. This provides overwhelming evidence against the null hypothesis. So, we must conclude that time has a significant, very strong effect on chicken mass. This is consistent with biological expectations: as chicks grow over time and their mass increase.
   - Note, however, that this model ignores diet, treats all chickens as a single population observed at different times. The significant result reflects that growth occurs over time but it does not tell us whether or how different diets contribute to that growth — only that time, on average, is strongly associated with (causing, in this instance) increasing body mass.

# 9. Correlation and association

! Self-assessment Task 9-1

For each of the following pairs of variables, decide which correlation method (Pearson, Spearman, or Kendall) is most appropriate and briefly justify your choice:

a. Body mass (g) and wing span (mm) in a sample of 120 birds — the scatterplot looks roughly linear. **(/2)**
b. Ecologist ranks 15 sites from least to most disturbed (1–15); another ecologist ranks the same 15 sites by species richness. The question is whether more disturbed sites tend to rank lower in diversity. **(/2)**
c. Sea surface temperature and chlorophyll-a concentration in a large oceanographic dataset — the relationship is clearly curvilinear (cooling increases phytoplankton). **(/2)**
d. For (c), does Spearman also have an advantage over Pearson when data are skewed? **(/2)**
e. A nutritionist scores diet quality on a 5-point ordinal scale and records BMI for 40 participants. **(/2)**

! Self-assessment Task 9-2

Find **two datasets** of your own and do a full correlation analysis on each. Briefly describe the data and why they exist. State the hypotheses, do an EDA, make exploratory figures, choose and justify the appropriate correlation method, assess assumptions, and write up the results in publication style.

**Rubric**

| Criterion | Excellent (Full Marks) | Partial Credit | Absent / Poor | Marks |
|---|---|---|---|---|
| **1. Dataset Choice and Justification** | Two variables (from one or more datasets) are clearly described and justified as candidates for correlation analysis; rationale is thoughtful and contextually informed. | Variables are chosen and described but the rationale is vague or unconvincing. | Variable selection appears arbitrary or trivial; little or no justification is given. | /2 |
| **2. Hypothesis Framing** | Null and alternative hypotheses are explicitly stated and aligned with the correlation analysis (*e.g.*, $H_0 : \rho = 0$). Contextual meaning is clearly explained. | Hypotheses are present but poorly articulated or lacking contextual relevance. | Hypotheses are missing, incorrect, or misaligned with the analysis. | /2 |
| **3. Exploratory Data Analysis** | EDA includes summary statistics, variable distribution inspection, and consideration of linearity or | EDA is attempted but lacks depth or overlooks important features such as skewness or re- | No meaningful EDA is performed before conducting the correlation. | /3 |
| **4. Method Selection and Justification** | The chosen correlation method is appropriate for the data and clearly justified; assumptions are explicitly addressed. | The method is reasonable but justification is weak or assumptions are not fully considered. | The method is inappropriate or unjustified; assumptions are ignored. | /3 |
| **5. Interpretation and Discussion** | Results are correctly interpreted in context; limitations and potential confounds are discussed; conclusions are appropriately cautious. | Interpretation is partially correct but overstates or understates the evidence; limitations are briefly mentioned. | Interpretation is incorrect or misleading; no discussion of limitations. | /3 |

A strong answer should include:

1. **Dataset and variable choice**: Two biologically meaningful variable pairs. Explain why the question is one of
2. **Hypotheses**:
3. **EDA and visualisation**:
4. **Correlation**:

Taken together, a submission shows that the student understands why correlation is appropriate, why a particular coefficient was chosen, and what the result does and does not mean biologically.

For rank-based methods, state the equivalent null for $\rho_s$ (e.g.

## 12. Simple linear regression

! Self-assessment Task 12-1

a. Examine the contents of the regression model object `sparrow_mod`. Explain the main components and how they relate to `summary(sparrow_mod)`. **(/3)**
b. Using values inside the model object, show how to reconstruct the observed response values from the fitted values and residuals. **(/3)**
c. Fit a linear regression through the model residuals and explain the result. **(/2)**
d. Fit a linear regression through the fitted values and explain the result. **(/2)**

**Answer**

a. **Contents of the model object**
   - ? (x 1) `sparrow_mod` is a list-like fitted model object, which can be inspected with `str(sparrow_mod)`.
   - ? (x 1) Its main components include coefficients, residuals, fitted values, the model frame, terms, contrasts, and other diagnostic quantities.
   - ? (x 1) `summary(sparrow_mod)` pulls out the most important inferential components from that object, including coefficient estimates, standard errors, test statistics, residual standard error, and model-fit summaries.

b. **Reconstructing the observed response**
   - ? (x 1) The key identity is: observed response = fitted value + residual.
   - ? (x 1) This can be reconstructed using the extractor functions:
   - ? (x 1) The same reconstruction can be done directly from components stored in the model object:

```
fitted(sparrow_mod) + resid(sparrow_mod)
```

```
sparrow_mod$fitted.values + sparrow_mod$residuals
```

c. **Regression of residuals on age**
   - ? (x 1) If the residuals are regressed on `age`, the fitted slope should be essentially zero and the test should be non-significant.
   - ? (x 1) This occurs because least-squares residuals are orthogonal to the predictor used in the original model fit.

d. **Regression of fitted values on age**
   - ? (x 1) If the fitted values are regressed on `age`, the fitted linear trend is recovered.
   - ? (x 1) The relationship is perfectly linear because the fitted values are generated directly from the regression model.

! Self-assessment Task 12-2

Complete a full simple linear regression analysis using life expectancy as the response and schooling as the predictor. Your job is not only to fit the model, but also to identify and justify the treatment of problematic observations.

Use the dataset kaggle_life_expectancy_data.csv.

You should do the following:

1. Prepare the data by selecting the variables needed for the analysis and removing rows with missing values.
2. Fit the initial simple linear regression model.
3. Plot the data and show the fitted regression line.
4. Check the initial model diagnostics graphically.
5. Identify the issue in the dataset.
6. Provide evidence for this issue as a table.
7. Explain why these cases appear problematic for this analysis.
8. Remove the problematic cases and refit the model.
9. Recheck the diagnostics graphically for the revised model.
10. Interpret the final model clearly and state whether it is an improvement over the initial fit. Tabulate the important model-fit statistics in support of your conclusion.
11. End with a short scientific write-up containing Methods, Results, and Discussion sections.

## Marking rubric

| Component | Marks |
|---|---|
| Data preparation | 3 |
| Initial model fitting and figure | 3 |
| Initial model diagnostics | 3 |
| Identifying and explaining the issue | 3 |
| Evidence for the issue presented as a table | 4 |
| Removing the problematic cases appropriately | 2 |
| Refitting the analysis | 3 |
| Rechecking the diagnostics | 3 |
| Interpreting the final model and comparing it with the initial model | 3 |
| Scientific write-up (Methods, Results, Discussion) | 3 |

**Total: 30 marks**

**Answer**



See the worked example here: kaggle_life_expectancy.qmd. Example marks allocations according to the rubric are assigned there.

# Bibliography